

JAMES RICKARDS

james.rickards@smu.ca
<https://jamesrickards-canada.github.io/>
<https://github.com/JamesRickards-Canada>

McNally North 120, Department of Mathematics and Computing Science
Saint Mary's University
Halifax, NS

POSITIONS

Associate Professor Saint Mary's University	Sept 2026 - present Halifax, NS
Assistant Professor Saint Mary's University	2024 - Aug 2026 Halifax, NS
Postdoctoral Fellow <i>Mentor: Katherine E. Stange</i> University of Colorado Boulder	2021 - 2024 Boulder, CO

EDUCATION

Doctor of Philosophy <i>Advisor: Henri Darmon</i> McGill University Thesis title: Intersections of closed geodesics on Shimura curves	2016 - 2021 Montreal, QC
Master of Arts Trinity College, University of Cambridge	2019 Cambridge, UK
Master of Mathematics Trinity College, University of Cambridge	2015 - 2016 Cambridge, UK
Bachelor of Arts (Hons) <i>Major: Mathematics</i> Trinity College, University of Cambridge	2012 - 2015 Cambridge, UK

RESEARCH INTERESTS

Computational number theory, algebraic number theory, circle packings, thin (semi)groups, arithmetic Fuchsian/Kleinian groups, binary quadratic forms, quaternion algebras, Shimura curves, continued fractions, visualization.

PUBLICATIONS AND PREPRINTS

Papers are ordered by the date the initial preprint was completed, as opposed to the final publication date.

11. Eisenstein circle packings and the Eisenpint Schmidt arrangement James Rickards, Katherine E. Stange Preprint	2026
10. Primes represented by shifted quadratic forms: on primitivity and congruence classes Elena Fuchs, Catherine Hsu, James Rickards, Damaris Schindler, Katherine E. Stange Acta Arithmetica 222 (2026), 371-391	2026
9. Prime and thickened prime components in Apollonian circle packings Elena Fuchs, Holley Friedlander, Piper Harris, Catherine Hsu, James Rickards, Katherine Sanden, Damaris Schindler, Katherine E. Stange Research Directions in Number Theory, Association for Women in Mathematics Series vol. 39 (2026)	2026
8. Reciprocity obstructions in semigroup orbits in $SL(2, \mathbb{Z})$ James Rickards, Katherine E. Stange Duke Mathematical Journal 174(15): 3197-3244 (15 October 2025)	2025
7. The local-global conjecture for Apollonian circle packings is false Summer Haag, Clyde Kertzer, James Rickards, Katherine E. Stange Annals of Mathematics (2) 200(2): 749-770 (September 2024)	2024

6. The Apollonian staircase	2024
James Rickards International Mathematics Research Notices, Volume 2024, Issue 2, January 2024, pp. 1340-1372	
5. Improved computation of fundamental domains for arithmetic Fuchsian groups	2022
James Rickards Mathematics of Computation 91 (2022), no. 338, pp. 2929-2954	
4. Hecke operators acting on optimal embeddings in indefinite quaternion algebras	2022
James Rickards Acta Arithmetica 204 (2022) no. 4, pp. 347-367	
3. Counting intersection numbers of closed geodesics on Shimura curves	2023
James Rickards Research in Number Theory 9 (2023), no. 2, Paper No. 20, 45 pp.	
2. Computing intersections of closed geodesics on the modular curve	2021
James Rickards Journal of Number Theory, 225 (2021), pp. 374-408	
1. When is a Polynomial a Composition of Other Polynomials?	2011
James Rickards American Mathematical Monthly, 118 (2011), no. 4, pp. 358-363	

RESEARCH GRANTS

Total funding for each grant is listed. All amounts are in CAD.

NSERC Discovery Grant	2025 - 2030
Thin groups, quaternion algebras, and computational number theory	\$187,500
FGSR Supplementary Funding for International Conference Participation for Faculty Members	2024
Arithmetic groups, hyperbolic manifolds and computation	\$1,000
FGSR University Grant in Aid of Research	2024 - 2025
Computational number theory	\$5,000
SMU Start-up Funding Package	2024 - 2027
	\$26,500

MEDIA

CU students follow their noses, disprove math conjecture	2023
Article about <i>The Local-Global Conjecture for Apollonian circle packings is false</i> Colorado Arts and Sciences Magazine, https://www.colorado.edu/asmagazine/2023/11/30/cu-students-follow-their-noses-disprove-math-conjecture	
The Hidden Connection That Changed Number Theory	2023
Contributed quotes Quanta Magazine, https://www.quantamagazine.org/the-hidden-connection-that-changed-number-theory-20231101/	
Two Students Unravel a Widely Believed Math Conjecture	2023
Article about <i>The Local-Global Conjecture for Apollonian circle packings is false</i> Quanta Magazine, https://www.quantamagazine.org/two-students-unravel-a-widely-believed-math-conjecture-20230810/	

CODE

Apollonian	PARI/GP
Computations for Apollonian circle packings, including basic operations, generating pictures in LaTeX, and a very efficient implementation for finding all missing curvatures up to a bound. Available at https://github.com/JamesRickards-Canada/Apollonian	
Apollonian-Prime	PARI/GP
Computations for thickened prime components of Apollonian circle packings, Available at https://github.com/JamesRickards-Canada/Apollonian-Prime	

Eisenstein

PARI/GP

Computations for Eisenstein circle packings, including basic operations, generating pictures in LaTeX, generating circle in the full Eisenstein/Eisenpint Schmidt arrangements, and computing all curvatures in a packing. Available at <https://github.com/JamesRickards-Canada/Eisenstein>

Fundamental domains for Shimura curves

PARI/GP

Computation of fundamental domains for arithmetic Fuchsian groups. Improves on the algorithms of Voight and Page, and is significantly more efficient than the live Magma implementation (from 100 to millions of times as fast, depending on the example). Will be integrated into PARI/GP.

Available at <https://github.com/JamesRickards-Canada/Fundamental-Domains-for-Shimura-curves>

Isogeny

PARI/GP, Sage

Computation of supersingular ℓ and L isogeny graphs, significantly more efficient than the live Sage implementation. Includes code to seamlessly use it inside of Sage.

Available at <https://github.com/JamesRickards-Canada/Isogeny>

Q-Quadratic

PARI/GP

Computing with integral binary quadratic forms and quaternion algebras over $\mathbb{Q}$. Includes algorithms to compute intersection numbers of modular geodesics, as described in my thesis and various papers.

Available at <https://github.com/JamesRickards-Canada/Q-Quadratic>

Semigroup Reciprocity

PARI/GP

Computation of orbits of semigroups, including efficient implementation of missing numbers in an orbit. This package accompanies the paper *Reciprocity obstructions in semigroup orbits in $SL(2, \mathbb{Z})$* , and includes methods to check various results.

Available at <https://github.com/JamesRickards-Canada/Semigroup-Reciprocity>

OTHER ACADEMIC WRITING

Competition Highlights: Canadian Mathematical Olympiad and Junior Olympiad (CMO/CJMO)

Paweł Prałat, James Rickards

Crux Mathematicorum: Vol. 50(8), October 2024; Vol. 51(7), September 2025

Running PARI/GP on Windows

Tutorial for installing and using PARI/GP on Windows computers.

Available at <https://pari.math.u-bordeaux.fr/PDF/PARIGP-Windows.pdf>

Polynomial Division in Number Theory

Crux Mathematicorum, Vol. 43(10), December 2017

Parametric Solutions to the Generalized Fermat Equation

Part III essay, Cambridge, 2016

Higher Power Reciprocity Laws

Rouse Ball Mathematical Essay, Cambridge, 2015

CONFERENCE TALKS

AMMCS-2026

Aug 2026

Eisenstein Circle Packings

Wilfrid Laurier University

ANTS XVII

Jul 2026

Continued fractions with the digits $\{1, 2\}$

Rijksuniversiteit Groningen

Atelier libpari 2025

Jun 2025

Supersingular Isogeny Graphs (in PARI/GP)

Université de Bordeaux

Arithmetic groups, hyperbolic manifolds and computation

Dec 2024

Totally geodesic surfaces in Bianchi orbifolds

Université de Bordeaux

ANTS XVI

Jul 2024

Reciprocity obstructions in continued fraction semigroups

MIT

Computational Aspects of Thin Groups

Jun 2024

The not-so-local-global conjecture

NUS

Renormalization, computation and visualization in Geometry, Number Theory and Dynamics	Sept 2023
The not-so-local-global conjecture	CIRM
LuCaNT	Jul 2023
Software demo: Computing fundamental domains for congruence arithmetic Fuchsian groups in PARI/GP	ICERM
Number Theory Informed by Computation	Aug 2022
Fast fundamental domains for arithmetic Fuchsian groups in PARI/GP	Park City Mathematics Institute
16th Atelier PARI/GP 2022	Jan 2022
Fundamental Domains for Shimura curves	U. Franche-Comté (participated online)
Lattices and Cohomology of Arithmetic Groups: Geometric and Computational Viewpoints	Oct 2021
Improved computation of fundamental domains for arithmetic Fuchsian groups	BIRS (online)
Front Range Number Theory Day	Sep 2021
Counting intersection numbers on Shimura curves	Colorado State University
Front Range Number Theory Day	Apr 2021
Fast computations of fundamental domains for Shimura curves	CU Boulder (online)
Quebec-Maine Number Theory Conference	Sep 2020
Computing with (indefinite) quadratic forms and quaternion algebras in PARI/GP	Laval University (online)
Quebec-Maine Number Theory Conference	Oct 2019
Intersection numbers of modular geodesics	University of Maine
Quebec-Maine Number Theory Conference	Oct 2018
Intersection numbers of modular geodesics	Laval University
CMS Summer Meeting	Jun 2018
Number theoretic intersection numbers on Riemann surfaces	University of New Brunswick
Montreal-Toronto Workshop in Number Theory	Dec 2016
Basic background on mock modular forms and weak harmonic Maass forms	University of Montreal

SEMINAR TALKS

The Ohio State University Number Theory Seminar	Apr 2025
Failure of the local-global conjecture in thin (semi)groups	The Ohio State University
Undergraduate Research Seminar	Oct 2024
Apollonian circle packings and the not-so-local-global conjecture	Saint Mary's University
Algebraic Geometry Seminar	May 2024
The not-so-local-global conjecture	UC Davis
PU/IAS Number Theory Seminar	Apr 2024
The not-so-local-global conjecture	Princeton University / IAS
Dalhousie Number Theory Seminar	Mar 2024
Quaternion algebras in number theory	Dalhousie University
Dalhousie Colloquium	Mar 2024
The not-so-local-global conjecture	Dalhousie University
Saint Mary's Colloquium	Jan 2024
Apollonian circle packings and thin groups	Saint Mary's University
Virtual Seminar on Geometry and Topology	Nov 2023
Failure of the local-global conjecture in thin (semi)groups	KIAS, South Korea
Penn State Algebra and Number Theory Seminar	Oct 2023
The not-so-local-global conjecture	Penn State
University of Washington Number Theory Seminar	Oct 2023
The not-so-local-global conjecture	University of Washington

Arithmetic Reflection Groups Seminar	Aug 2023
The not-so-local-global conjecture	Online
Five College Number Theory Seminar	Nov 2022
The Apollonian Staircase	Amherst College
Brown University Algebra and Algebraic Geometry Seminars	Nov 2022
The Apollonian Staircase	Brown University
International Seminar on Automorphic Forms	May 2021
Counting intersection numbers on Shimura curves	TU Darmstadt/ETH Zurich (online)
Rutgers Number Theory Seminar	Oct 2019
Intersection numbers of modular geodesics	Rutgers University
Laval Number Theory Seminar	Oct 2019
Intersection numbers of modular geodesics	Laval University

TEACHING EXPERIENCE - SAINT MARY'S UNIVERSITY

Math 2305 <i>Survey of Discrete Mathematics</i>	F24 + 2R, F25 + 1R, F26 + 2R
Math 2310 <i>Introductory Analysis</i>	W25 + 1R
Math 3025* <i>Introduction to Number Theory</i>	W25, F26
Math 3441 <i>Real Analysis I</i>	F25
Math 4420 <i>Abstract Algebra I</i>	W26
Math 4426 <i>Theory of Functions of a Complex Variable I</i>	W27

* Math 3025 was taught once as Math 3827 (Special Topic: Number Theory) before being added as a permanent course.

TEACHING EXPERIENCE - UNIVERSITY OF COLORADO BOULDER

Math 2001 <i>Introduction to Discrete Mathematics</i>	Fall 2022 - 2 sections, Spring 2024
Math 2130 <i>Linear Algebra for Non-Math Majors</i>	Fall 2021, Spring 2022
Math 3001 <i>Analysis 1</i>	Fall 2023
Math 3110 <i>Introduction to the Theory of Numbers</i>	Spring 2022, Spring 2024
Math 8174 <i>Topics in Algebra - Quaternion Algebras (Graduate course)</i>	Spring 2023

TEACHING EXPERIENCE - OTHER

TA for PCMI graduate course	Summer 2022
TA for Jan Vonk's one week long course at the Park City Mathematics Institute graduate summer school	
Math 141 TA <i>Integral Calculus</i>	Fall 2017, Fall 2018
McGill University	

MENTORSHIP

Summer Research - Saint Mary's

Advised summer undergraduate research students for 12-16 weeks on projects involving Apollonian circle packings. In 2025, advised 3 students. In 2026, advised 3 students.

Honours Thesis Advisor

Co-advisor to Clyde Kertzer on symmetries in Apollonian circle packings (Fall 2023 - Spring 2024).

2023 REU - CU Boulder

Ran an REU jointly with Katherine E. Stange on Apollonian circle packings. Supervised one undergraduate student (Clyde Kertzer) and one first year graduate student (Summer Haag).

Math camp leader and trainer

2015, 2017 - 2019

Mentored and trained Canadian high school students interested in contest math at four (week-long) IMO (International Mathematical Olympiad) winter camps, as well as four IMO summer camps (3 weeks long each), and one EGMO (European Girls Mathematical Olympiad) training camp (week-end).

SCHOLARSHIPS

Vanier Canada Graduate Scholarship \$50,000 CAD/year	2018 - 2021
NSERC CGS D	2018 (Declined)
Schulich Fellowship <i>McGill University</i> \$25,000 CAD/year	2016 - 2018
Trinity College Woods Scholarship \$25,000 CAD/year	2015 - 2016
Cambridge Trusts Scholarship \$25,000 CAD/year	2015 - 2016
Blyth Cambridge Commonwealth Scholarship \$50,000 CAD/year	2012 - 2015
Lazaridis Olympiad Scholarship to University of Waterloo	2012 (Declined)

MATH HUSKIES

Putnam Mathematics Competition Organize participation of Saint Mary's team in the Putnam mathematics contest.	Fall 2025 - present
Science Atlantic Organize participation of Saint Mary's team in the Science Atlantic mathematics contest.	Fall 2024 - present
Weekly training sessions Run weekly training sessions during Fall semester preparing Saint Mary's University students for the Science Atlantic and Putnam math contests.	Fall 2024 - present

CANADIAN MATHEMATICAL SOCIETY SERVICE

Canadian International Mathematical Olympiad Committee chair Organize a 1 week training camp every January for 15-20 students, coordinate the writing and grading of the APMO contest for around 30 students, organize the team selection test and grading for 12-18 students, help plan the 2 week summer training camp and the IMO itself, plus general administrative duties.	2019 - present
Canadian Junior Mathematical Olympiad coordinator This is a variant of the CMO for students in grade ≤ 10 . I am in charge of ensuring that problems get selected for the contest.	2019 - present
Canadian International Mathematical Olympiad Committee member Propose problems and participate in grading. This is a high school olympiad style contest for 70-100 Canadian students.	2016 - present
Canadian Open Mathematics Challenge Problems Committee member Proposed problems and participated in the selection of the final contest.	2013 - 2021

INTERNATIONAL MATHEMATICAL OLYMPIAD SERVICE

Team Canada Leader Observer Help run the 1 week long winter training camp and 2 week long summer training camp, then head to the IMO early to participate in problem selection and grading of our student's problems.	2019
Team Canada Leader The leader and leader observer have similar duties, but the leader is officially in charge.	2017, 2018
Team Canada Deputy Leader Observer Similar duties to the leader, except the deputy leader and observer accompany the students to the IMO. They still participate in grading/coordination of the Canadian student submissions.	2015

OTHER MATHEMATICAL OLYMPIAD SERVICE

Olympiade Francophone de Mathématiques

Organizer for the Canadian team

2021 - present

PAPER REVIEW

Reviewed papers/project proposals for (among others): Acta Arithmetica, Banff International Research Station, Commentarii Mathematici Helvetici, Communications in Algebra, International Mathematical Research Notices, Journal of Number Theory, Journal of the European Mathematical Society, and Transactions of the American Mathematical Society.

OTHER SERVICE

Undergraduate thesis reviewer

Served as the reader for one undergraduate thesis.

2026 - present

Saint Mary's

Code Huskies

Gave a couple of training sessions each semester for the SMU Code Huskies, an undergraduate programming contest club.

2025 - present

Saint Mary's

Comprehensive exams

Committee member for three comprehensive oral exams at CU Boulder.

2021 - 2024

CU Boulder

SKILLS

Languages: English (native), French (limited working proficiency)

Programming:

- High proficiency: C, PARI/GP
- Medium proficiency: Python
- Some familiarity: HTML, Magma, Mathematica, SageMath